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Peak selection matters in principal component analysis: A case study of syntrophic microbes

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In situ liquid time-of-flight secondary ion mass spectrometry (ToF-SIMS) is a powerful technique to study surface characterization of living biofilms in hydrated conditions. However, ToF-SIMS data analysis is still a great challenge in complicated bacterial biofilms, because many interference peaks from the medium may result in inaccurate interpretation. In this study, two syntrophic *Geobacter* populations are investigated using *in situ* liquid ToF-SIMS to reveal the biofilm surface changes between them due to direct interspecies electron transfer. By comparing spectral principal component analysis (PCA) results of all peaks and selected peaks, the authors find that spectral peak overlay is an effective strategy to reduce the matrix effect in handling complex ToF-SIMS data. Additionally, the spectral PCA results of high intensity and high resolution data obtained from liquid ToF-SIMS are compared. Selected peaks, amino acid peaks, and water cluster peaks spectral PCA produce nice separation among samples in both high intensity and high resolution data sets. However, the high resolution data show better separation between coculture planktonic and coculture aggregates, confirming that the higher mass accuracy is useful in the analysis of microbial samples. In conclusion, the results show that peak selection is critical for acquiring effective microbial information and interpretation of syntrophic *Geobacter* using spectral data from *in situ* liquid ToF-SIMS. *Published by the AVS.* https://doi.org/10.1116/1.5118237

I. INTRODUCTION

Geobacter species are the most widely distributed iron-reducing bacteria and can reduce iron Fe (III) into Fe (II) and facilitate coupling of carbon, nitrogen, and phosphorus cycles in the natural anaerobic environment.[1,2] Therefore, the redox process mediated by *Geobacter* is important in the biogeochemical cycle. *Geobacter sulfurreducens* and *Geobacter metallireducens* were reported to have the potential of extracellular electron exchange, and they were deemed efficient in the application of bioremediation by reducing various organic pollutants and toxic heavy metals.[3,4] Previous studies show that the cooperation of the two *Geobacter* species through direct interspecies electron transfer (DIET) can contribute to the syntrophic metabolism, regulate their behavior, and optimize the performance in the engineered systems or in natural microbial communities.[5,6] The presence of small Fe-containing particles on the surface and microbial nanowires of the *Geobacter sulfurreducens* are considered responsible for extracellular electron transfer.[7] The role of conductive nanowires in DIET of *Geobacter* species can be promoted by the addition of conductive nanoparticle materials.[8,9] However, the details of chemical composition and structure properties on the syntrophic *Geobacter* biofilm surface remain unclear.

Time-of-flight secondary ion mass spectrometry (ToF-SIMS) is a widely used imaging mass spectrometry technique, and it offers surface chemical analysis and chemical imaging with nanoscale details in various materials.[10–14] Compared to other imaging mass spectrometry techniques, such as desorption electrospray ionization and matrix-assisted laser desorption ionization, ToF-SIMS uses high energy primary ions or ion clusters (i.e., Bi_3^+, Au_3^+, and C_{60}) and offers the highest lateral resolution in the submicrometer range.[15,16] Previously, ToF-SIMS has been used to analyze biological materials, extracellular polymeric substance (EPS) of bacteria and microalgae, including protein, lipid, amino acid, and many organic molecules.[10,11,17–20]

However, biological sample preparation for ToF-SIMS normally requires freezing the biomaterials under −80 °C or drying them with air or nitrogen prior to ToF-SIMS analysis in vacuum.[18] We developed a vacuum compatible microfluidic reactor, system for analysis at the liquid vacuum interface (SALVI), to enable liquid analysis using ToF-SIMS that has been traditionally limited to solid samples. Liquid SIMS was developed as a new method to study bacterial biofilms *in vivo* in their hydrated state and prevent damage or change of the biological system during sample preparation. We have shown recently that a variety of bacteria biofilms can be cultured in the SALVI microreactor and imaged using optical microscopy (e.g., confocal laser scanning microscopy and structured illumination microscopy) and nuclear magnetic resonance imaging besides *in situ* liquid SIMS.[19,21–24]

ToF-SIMS provides big data with hundreds of peaks in every spectrum for every pixel, and the intensity of each peak can vary due to the composition, structure, order, and orientation of the surface species.[17] Whether it is liquid SIMS or static SIMS analysis, we face challenges to process

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a large amount of data and determine the changes or differences of the spectral features among samples. Principal component analysis (PCA) is a widely used multivariate analysis method to process ToF-SIMS data,[25,26] because it can help identify discriminations and similarities among samples and determine main components compared to spectral analysis solely. However, PCA results may be influenced by the interference peaks due to their high intensity values, some of which arise from the matrix effect.[13] It is well known that peak selection is the most important data processing step prior to PCA.[13] Specifically for microbial cell samples, the culture medium contains rich nutrients or even organic matters and the matrix effects are rather significant.[27]

We used two medium solutions to culture the syntrophic *Geobacter* community *Geobacter sulfurreducens* DL1 (ATCC 51573) and *Geobacter metallireducens* GS15, namely, neutral buffered fumarate (NBF) and Fe (III)-citrate Fresh Water (FC). Both showed the nutrient peaks in comparison with bacteria samples [Fig. S1 (Ref. 43)].[28] Except removing the known interference peaks that are typical of the polydimethyl siloxane (PDMS) device and the primary bismuth ion peaks from the primary ion beam,[21] manual peak selection is strongly recommended in complex organic and biological surfaces in liquid SIMS analysis.[13] Although manual peak selection may be time consuming, it is considered a reliable way to ensure that the spectral features of the importance of the sample and peaks observed are placed properly among all samples.[25,26]

In this work, we propose the following two hypotheses concerning data treatment and selection in the syntrophic *Geobacter* system studied using *in situ* liquid SIMS. First, the matrix effect hinders the interpretation of microbial interactions. We adopt the spectral overlay method to reduce the matrix effect by comparing spectral PCA results of all peaks and selected peaks. Second, both high resolution and high intensity liquid SIMS data are suitable for data analysis. According to our earlier work, we obtain high intensity data and high resolution data using different pulse widths during the microbial biofilm analysis using liquid ToF-SIMS. The higher intensity data have high spatial resolution, and they are normally used for image analysis. As the depth profile continues, a pulse with a narrower width is used, and lower counts of ions are observed. These data have slightly higher mass resolution and are normally used for spectral analysis.[19,29] A schematic is depicted in Fig. S2.[43] Amino acid, main components of bacterial secretion, and water cluster, an important medium for living biofilms, are chosen for more analysis. We compare spectral PCA results based on the spectral overlay peak selection between the high intensity and the high resolution segment within liquid SIMS acquisition to determine whether the two sets of data are suitable for spectral analysis.

II. EXPERIMENT

A. SALVI

SALVI is a microfluidic device developed for *in situ* liquid analysis in vacuum platforms. The fabrication details have been reported in our previous publications.[30,31]

B. Bacterial culture

Geobacter sulfurreducens DL1 (ATCC 51573) and *Geobacter metallireducens* GS15 (ATCC 53774) were obtained previously.[32,33] Similar to our past reports,[28] *G. sulfurreducens* DL1 was cultured with NBF medium, and *G. metallireducens* GS15 was cultured with FC medium at 30 °C in a dark incubator. *G. sulfurreducens* DL1 and *G. metallireducens* GS15 were cultured and reached the exponential phase for the follow on experiment. For planktonic bacteria samples, pure cultured *G. sulfurreducens* DL1 or *G. metallireducens* GS15 cells were injected into the SALVI device separately and immediately brought to the ToF-SIMS instrument for *in situ* analysis. The cocultured planktonic bacteria were cultured with NBF medium by inoculating 1% *G. sulfurreducens* DL1 and 1% *G. metallireducens* GS15 in anaerobic tubes. After approximately 35 days, the cocultured planktonic bacteria were transferred into the SALVI device for liquid SIMS analysis.

For bacteria biofilm samples, *G. sulfurreducens* DL1 and *G. metallireducens* GS15 were directly cultured in SALVI cells. Briefly, the prepared bacteria cultures (i.e., 1% *G. sulfurreducens* DL1, 1% *G. metallireducens* GS15, or cocultures including 1% *G. sulfurreducens* DL1 and 1% *G. metallireducens* GS15) were inoculated into SALVI devices and cultured continuously with fresh medium solution for bacterial biofilm growth. Pure cultured DL1 and GS15 biofilms were formed around 5–7 days; coculture aggregates were formed around 45 days. The individual NBF and FC medium were also analyzed as medium controls.

C. Liquid ToF-SIMS

Briefly, a ToF-SIMS V spectrometer (IONTOF GmbH, Münster, Germany) was used. For liquid SIMS, the data acquisition process is depicted in Fig. S2.[43] The ion intensity of liquid molecules remains low before Bi_3^+ punched through the SiN membrane of SALVI (Region I). As the SiN membrane is drilled through, the ion intensity gradually increases (Region II) and reaches a stable condition of the high intensity of spectra (Region III). As the depth profile increases, a narrower pulse is used resulting in lower ion intensities of the ions of interest in the liquid (Region IV).[19,21,24,29]

D. PCA data analysis

The liquid SIMS data are mass calibrated and reconstructed using the ion-tof software (Surface Lab, version 6.4). More details are seen in our previous reports.[21,29,34] After constructing spectral data, the mass spectra are calibrated using Si^+, Bi^+, Bi_2^+, and Bi_3^+ in the positive ion mode and Si^-, SiO_2^-, PO_2^-, PO_3^-, $Si_2HO_5^-$, and $Si_3HO_7^-$ in the negative ion mode, respectively. We checked the measurement reproducibility for every sample (Fig. S3).[43] The calibrated mass spectra are compared and analyzed further with PCA.

We first normalized the calibrated spectra to the total ion intensity and then performed square root transformation and mean centering before running spectral PCA using the MATLAB software (R2012a, Math Works, Inc., MA). The

corresponding scores plots and loadings plots were plotted using ORIGIN.

E. Peak selection

The base peak list comes from the ToF-SIMS analysis and constructed directly by using the ION-TOF software. More details can be seen in our previous reports.[21,29,34] Prior to spectral PCA, the known systematic interference peaks such as PDMS (e.g., m/z$^+$ 147 $Si_2OC_5H_{15}^+$, 221 $Si_3O_2C_7H_{21}^+$, 281 $Si_4O_4C_7H_{21}^+$, 295 $Si_4O_3C_9H_{27}^+$, and 369 $Si_5O_4C_{11}H_{33}^+$)[21] and Bi primary ions (e.g., m/z$^+$ 209, 418, and 657) were excluded. Other peaks excluding these systematic interference peaks are termed as "all peak" for simplicity. In spite of removing the obvious interference peaks from the SALVI device and the primary ion beam, additional substantial interference peaks were found from the medium salts. Their intensities were high and potentially buried the microbial peaks of interest to the syntrophic community [Fig. S1 (Ref. 43)].[28]

To reduce the matrix effect, the spectral overlay strategy was employed to reduce the medium matrix effect in peak selection. Here, these peaks reducing the matrix effect are named "selected peak." First, each spectrum was normalized to its selected total ion intensity after removing the apparent interference peaks mentioned above. The spectral matrix across all samples (e.g., coculture aggregate, coculture planktonic, DL1 biofilm, DL1 planktonic, GS15 biofilm, GS15 planktonic, NBF medium, and FC medium) are displayed along the same coordinate axis (Fig. 1). Large potential interference peaks were removed according to the following procedure: (1) we first select the maximal values among the samples for each unit mass and (2) if the normalized ion intensity of the medium sample captures over 90% of the maximum value obtained from step (1) and they are mostly dominant in the medium solution spectra, we would remove this particular peak from the peak list. Thus, potential interference peaks could be removed from the peak list. Further, we perform PCA using the reduced peak list obtained from step (2). (3) By evaluating the scores plots and loadings plots, peaks that contributed highly to the medium samples were excluded from the peak list. Finally, potential characteristic peaks, including amino acids, lipids, water clusters, and fatty acids, which are closely related to bacterial biofilm components were kept for additional analysis. The process for selecting peaks and data analysis is shown in Fig. 2.

III. RESULTS AND DISCUSSION

Spectral PCA results can be influenced by interference peaks when analyzing ToF-SIMS data, this is especially true from the culture medium in complicated microbial systems.[13] Therefore, peak selection is a known challenge for obtaining insight into the true microbial characteristics. Here, we employ the spectral peak overlay strategy to reduce the interference from medium salts and focus on the comparison of ToF-SIMS spectral PCA results before and after peak selection using a spectral overlay of six different types of *Geobacter* samples and two kinds of culture medium solutions (Fig. 1). Furthermore, we discuss the application of high intensity data and high resolution data obtained from *in situ* liquid SIMS in spectral PCA.

A. Spectral overlay to remove interference peaks from the medium: Comparison of spectral PCA between all peak and selected peak

To get a full appreciation of the effect of the spectral overlay, spectral PCA of all peak was compared with that of the selected peak reducing the matrix effect. Whether for all peak spectral PCA or selected peak, the two *Geobacter* populations can be clearly separated [Figs. 3 and S4 (Ref. 43)] in PC1 and PC2, respectively. PC1 and PC2 explain 90.06% of the results for all peak spectral PCA and 72.67% of results for selected peak spectral PCA. Other PCs (i.e., PC3, PC4, and PC5) are not shown, as they are not able to separate samples well. When only removing the system interference peaks, nutrient medium samples occupy the highest scores values in positive PC1 and PC2 for all peak spectral PCA, and the dominance of these medium peaks interferes with the interpretation of other peaks characteristic of each *Geobacter* population. Several peaks with high scores on PC1, such as m/z$^+$ 39 K^+ and 57 $Ca(OH)^+$ might come from NBF, and m/z$^+$ 56 Fe^+ from FC. The more interesting peaks,

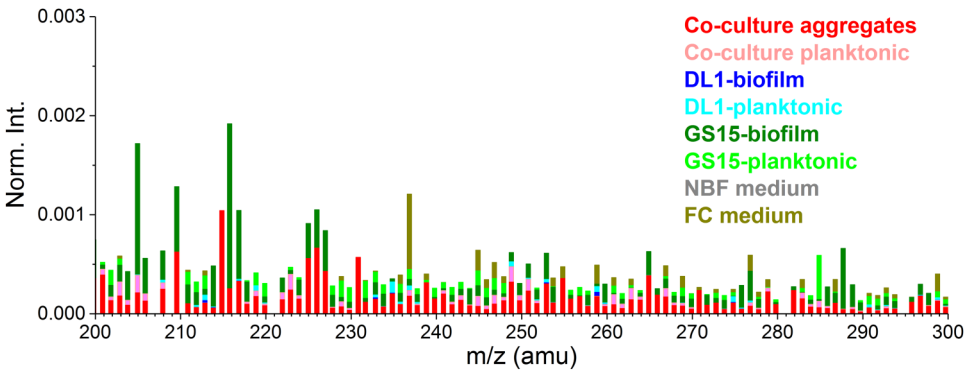


FIG. 1. Spectral overlay of representative positive spectra of eight samples in the range of m/z$^+$ 200–300. Norm. Int. stands for normalized intensity.

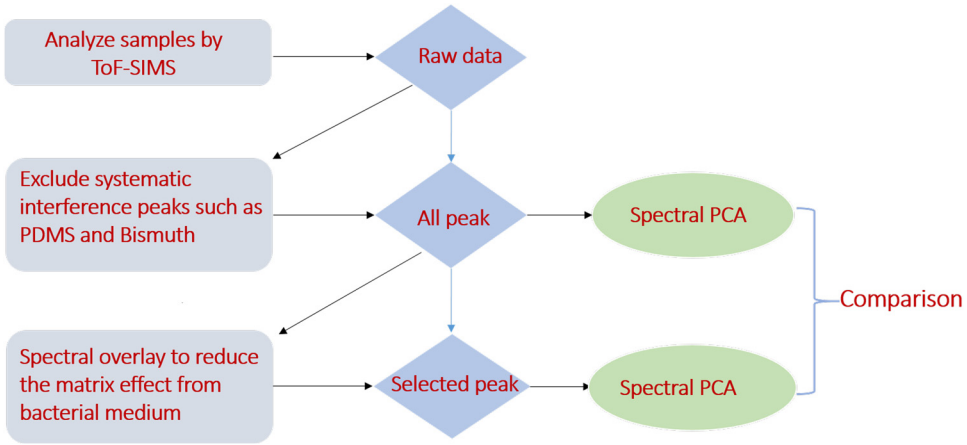


FIG. 2. Schematic showing the process of peak selection.

such as m/z$^+$ 159 $C_6H_7O_5^+$ polymer and m/z$^+$ 175 $C_{10}H_9NO_2^+$ quinolone, exist in coculture samples and *Geobacter* DL1. Figure 3(a) shows that PC1 separates the GS15 and DL1, GS15 and coculture samples. Combined with the loadings plot in Fig. 3(b), a series of amino acid peaks (m/z$^+$ 30 CH_4N^+ Gly, 44 $C_2H_6N^+$ Ala, 68 $C_4H_6N^+$ Pro/Lys, 70 $C_3H_3NO^+$ Asn, 72 $C_4H_{10}N^+$ Val, 82 $C_4H_6N_2^+$ His, 83 $C_5H_7O^+$ Val/Glu, 98 $C_4H_4NO_2^+$ Asn, 100 $C_4H_{10}N_3^+$ Arg, 112 $C_5H_8N_3^+$ Arg, 120 $C_8H_{10}N^+$ Phe)[21,28] are determined to contribute to the observed separation, which indicates that amino acid components of GS15 are different from those of DL15 and coculture samples [Figs. 3(a) and 3(b)]. According to the PC2 scores and loadings of all peak PCA in the positive ion mode [Figs. S4(a) and S4(b) (Ref. 43)],

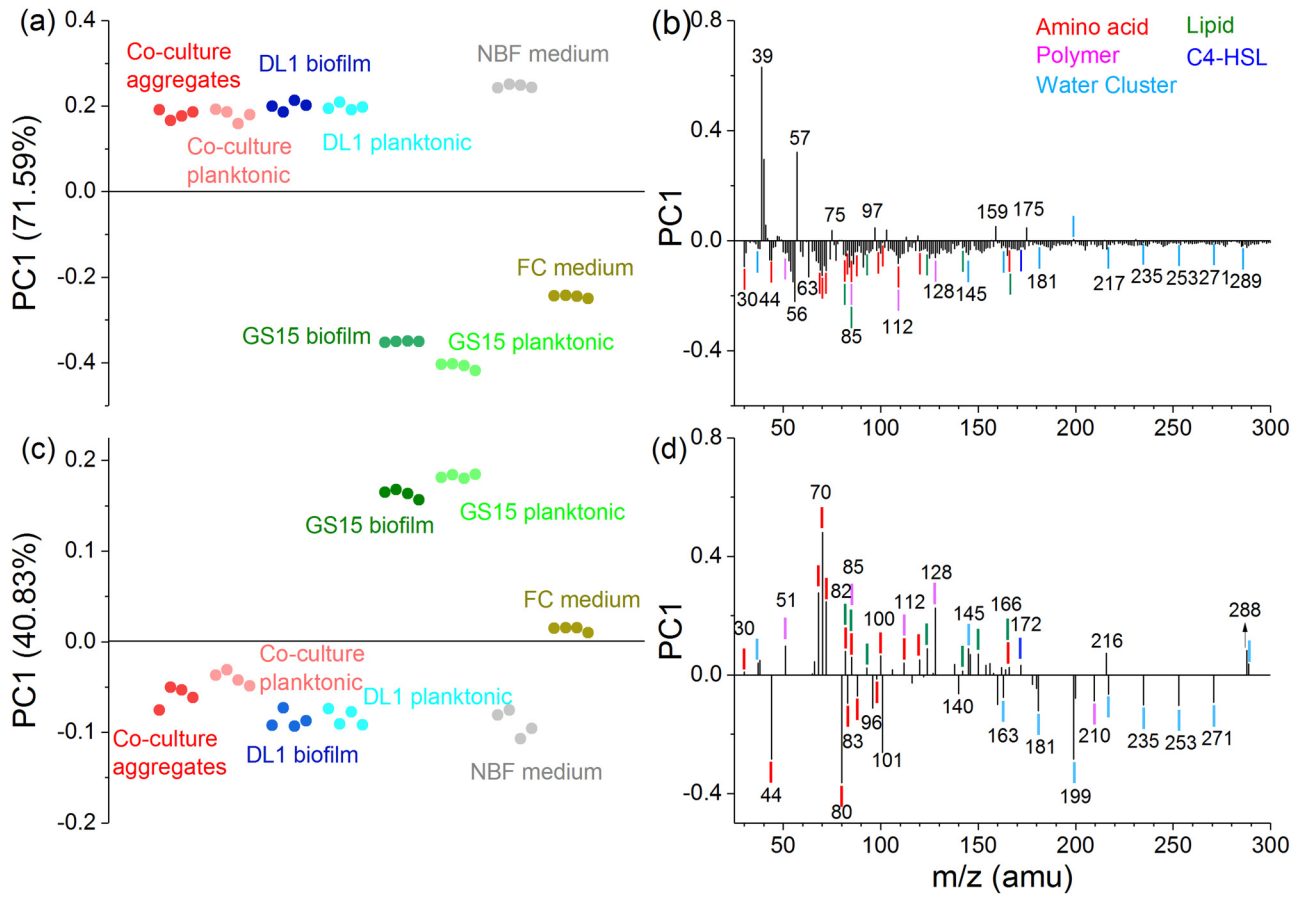


FIG. 3. Comparison of all peak and selected peak spectral PCA in the positive mode. PC1 scores plot (a) and loadings plots (b) of all peak PCA; PC1 scores plot (c) and loadings plot (d) of selected peak PCA using the spectral overlay to reduce the matrix effect.

m/z$^+$ 41 $C_3H_5^+$ (PDMS), 59 CH_3SiO^+ (PDMS), and other peaks representative of the FC medium show high loadings values. Amino acid peaks are also found in PC2 loadings. Similarly, characteristic lipid peaks are found in the all peak PCA PC2 loadings plot, for example, m/z$^+$ 82 $C_4H_6N_2^+$/$C_5H_8N^+$ histidine/phosphatidylcholine, 93 $C_7H_9^+$ phosphatidylcholine, 124 $C_2H_7NPO_3^+$ GPEtn, 142 $C_2H_9NPO_4^-$ DPPE headgroup fragment, and 166 $C_5H_{13}NPO_3^+$/$C_9H_{12}NO_2^+$ phosphatidylcholine or phenylalanine.[21,28] Peaks such as m/z$^+$ 51 $C_4H_3^+$, 85 $C_6H_{13}^+$, 112 $C_6H_{10}NO^+$, and 128 $C_{10}H_8^+$ are regarded as polymers. The peak m/z$^+$ 85 might also be attributed to phosphatidylcholine/glycine. Many water cluster peaks are found in *Geobacter* samples.

Additional spectral PCA results using selected peaks based on reducing the matrix effect are depicted in Figs. 3(c), 3(d), S4(c), and S4(d) (Ref. 43) in the positive mode. All samples were well discriminated in PC1 (40.83%) and PC2 (31.84%). The score values of the FC medium sample are largely reduced in PC1 [Fig. 3(c)] and the NBF medium sample in PC2 [Fig. S5(c) (Ref. 43)]. Amino acid, lipid, polymer, water clusters, and N-butanoyl-L-homoserine lactone peaks have higher loadings in the selected peak PCA results. Although the separation among samples is similar, most microbial component peaks have higher loading values in the spectral PCA using spectral overlay peak selection in comparison with that excluding system interference peaks. This helps elucidate the microbial significance when interpreting the PCA results.

Similar findings are found in all peak and selected peak spectral PCA plots in the negative ion mode [Figs. S5 and S6 (Ref. 43)]. Although PC1 scores and loadings plots still suggest some influence from the medium solutions, PC2 results are more satisfactory. Many interesting microbial components contribute to the separation in the PCA results, such as lipid, fatty acid, or DNA base. All peak PCA analysis shows high contributions from medium peaks and much lower loading values for peaks with microbial importance. Looking into the selected peak PCA, EPS peaks including lipid (m/z$^-$ 126 $C_2H_9NPO_3^-$ 2-aminoethylphosphono lipids and 196 $C_5H_{11}NPO_5^-$ phosphatidylethanol-amine), fatty acid (m/z$^-$ 223 $C_{14}H_{23}O_2^-$ phosphoinositol or phosphatidylinositol and 239 $C_{15}H_{27}O_2^-$ aliphatic chains of fatty acid), and DNA constituents (m/z$^-$ 110 $C_4H_4N_3O^-$ Cyt-H DNA base and 150 $C_5H_4N_5O^-$ Gua-H DNA base) have higher loadings values in PC1 and PC2, which would help us obtain a better insight into the surface changes of bacterial cells, biofilms, and the cocultured community. Additional peak identifications in the positive and negative mode are provided in Tables S1 and S2.[43]

EPS is the main component of bacterial biofilm due to bacterial secretion. EPS is complex and mainly includes proteins, polysaccharides, lipids, and nucleic acids.[35–37] Our results show an abundance of peaks of amino acids, lipids, and other small organic molecules. We also observe changes among them corresponding to the state of single populations or the cocultured community. Our previous reports of syntrophic *Geobacter* analyzed in dried samples showed that proteins representative of pili as well as fatty acids and lipid contents are strongly associated with the aggregates.[28]

These results give us good guidance to detect the surface chemical and molecular constitution in a complex microbial system, especially for the bacterial cells which need many medium salts to grow. The spectral overlay method is effective to exclude the matrix effect for peak selection. Our results show that the main EPS components, such as amino acids, lipids, or fatty acids, can distinguish different *Geobacter* species and the cocultured community. Moreover, the findings are better presented in the PCA using selected peaks after reducing the matrix effect.

B. The effects of mass accuracy and resolution on spectral PCA: Comparison of high intensity and high resolution spectral PCA

The experimental conditions of liquid ToF-SIMS have been optimized in our previous work.[19,21,29] The high intensity data and high mass resolution data cannot be achieved simultaneously, and the depth profile of Bi_3^+ going through the SiN membrane of SALVI and the biofilm was also verified by atomic force microscopy.[19] When the SiN membrane was drilled through, high intensity data were achieved. At the last part of data acquisition, a narrower pulse width was used for high resolution data with high accuracy [Fig. S2 (Ref. 43)]. However, whether the two sets of data are significantly different in spectral analysis remains unclarified. In this work, to ascertain the difference of spectral PCA of high intensity data and high resolution liquid SIMS data, we evaluate results in the following: (1) selected peak by using spectral overlay to reduce the effect of medium interference peaks; (2) selected amino acid peaks; and (3) selected water cluster peaks.

The selected peaks of high intensity and high resolution data in the positive mode were analyzed by spectral PCA as an example in Fig. 4. Figures 4(a) and 4(b) represent PC2 scores and loadings plots of high intensity data, respectively, and Figs. 4(c) and 4(d) high resolution data. We find that the microbial and medium samples are separated significantly. Coculture planktonic and coculture aggregates are distinguished better in high resolution data than high intensity data according to the PC2 score values [Figs. 4(a) and 4(c)]. The GS15 biofilm and the GS15 planktonic cell can be also well discriminated by the PC2 scores plot in the positive mode. The corresponding loadings plots in PC2 of high intensity and high resolution data show similar results, and peaks characteristic of microbes, such as amino acid peaks 68 $C_4H_6N^+$ Pro/Lys, 70 $C_3H_3NO^+$ Asn, and 72 $C_4H_{10}N^+$ Val, are highlighted in both high intensity and high resolution spectral PCA. PC1 results of high intensity and high resolution data spectral PCA in the positive mode are shown in Fig. S7 (Ref. 43). All biological and medium samples are distinguished in the PC1 scores plot, especially GS15 and DL1 [Figs. S7a and S7c (Ref. 43)]; however, PC1 results are not better than PC2 results concerning coculture samples.

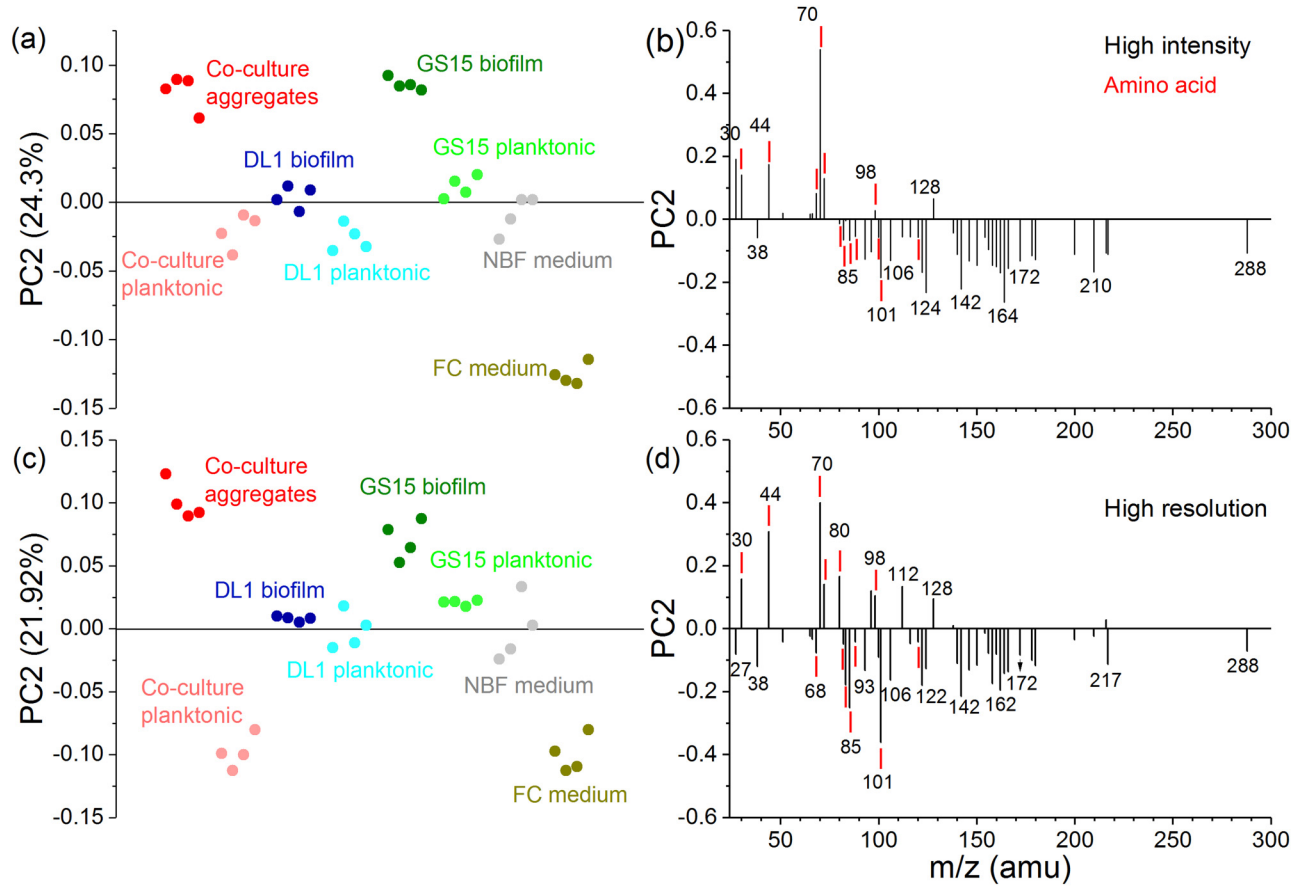


FIG. 4. Comparison of selected peak spectral PCA between high intensity and high resolution data in the positive mode. (a) PC2 scores plot of high intensity data; (b) PC2 loadings plot of high intensity data; (c) PC2 scores plot of high resolution data; and (d) PC2 loadings plot of high resolution data.

The latter is considered to be more significantly related to DIET between syntrophic *Geobacter* species.[28] Similar loading plots' results are obtained for both high intensity and high resolution data and similar microbial peaks (e.g., amino acid peaks) are shown with high loadings in PC1 [Figs. S7(b) and S7(d) (Ref. 43)]. These results illustrate that both high intensity data and high resolution data are suitable for spectral analysis. These findings indicate that bacterial metabolite changes result in the surface composition changes in the biofilm in comparison with planktonic cultures. Our recent dry ToF-SIMS analysis for this *Geobacter* syntrophic system also show that the biofilm composition changes adapted to the microbial growth condition and environment, and many amino acid, lipids, fatty acid peaks are observed, representative of the microbial system being studied.[28] Bacteria can coordinate their system-level behavior in a biofilm, for example, in growth dynamics. Liu *et al.*[38] has reported that distant *Bacillus subtilis* biofilms can also communicate by electricity signals to modulate their growth and metabolism in the syntrophic community.

C. The role of amino acids in cocultured aggregates

Furthermore, we select amino acid peaks as an example to compare the spectral PCA results between high intensity and high resolution data in the positive mode (Fig. 5), because amino acids are fragments of proteins. Proteins play an important role in DIET as pilis in GS15.[39,40] The scores plot shows that both GS15 biofilm and GS15 planktonic, coculture aggregates and coculture planktonic cultures are significantly separated in PC2; however, the spectral PCA of the high resolution data has a better discrimination of coculture planktonic and coculture aggregates [Figs. 5(a) and 5(c)]. The characteristic amino acid peaks m/z$^+$ 30 CH_4N^+ Gly, 44 $C_2H_6N^+$ Ala, and 70 $C_4H_8N^+$ Pro are used to distinguish the coculture aggregates, GS15 biofilm and their corresponding planktonic cells. Amino acid peaks m/z$^+$ 68 $C_4H_6N^+$ Pro/Lys, 82 $C_4H_6N_2^+$ His, 83 $C_5H_7O^+$ Val/Glu, 88 $C_3H_6NO_2^+$ Asn/Asp, 100 $C_4H_{10}N_3^+$ Arg, 101 $C_4H_{11}N_3^+$ Arg, and 120 $C_8H_{10}N^+$ Phe mainly come from planktonic samples based on the PC2 loadings plot. PC1 results in the positive mode also discriminate *Geobacter* samples and medium, and the loading plots are similar. However, score values show that *Geobacter* samples are not well separated in PC1 for both high intensity and high resolution data spectral PCA [Fig. S8 (Ref. 43)]. These results indicate that the changes of amino acid fragments reveal the role of proteins in biofilms and aggregates from liquid SIMS observations, in agreement with previous findings.[28] In this work, amino

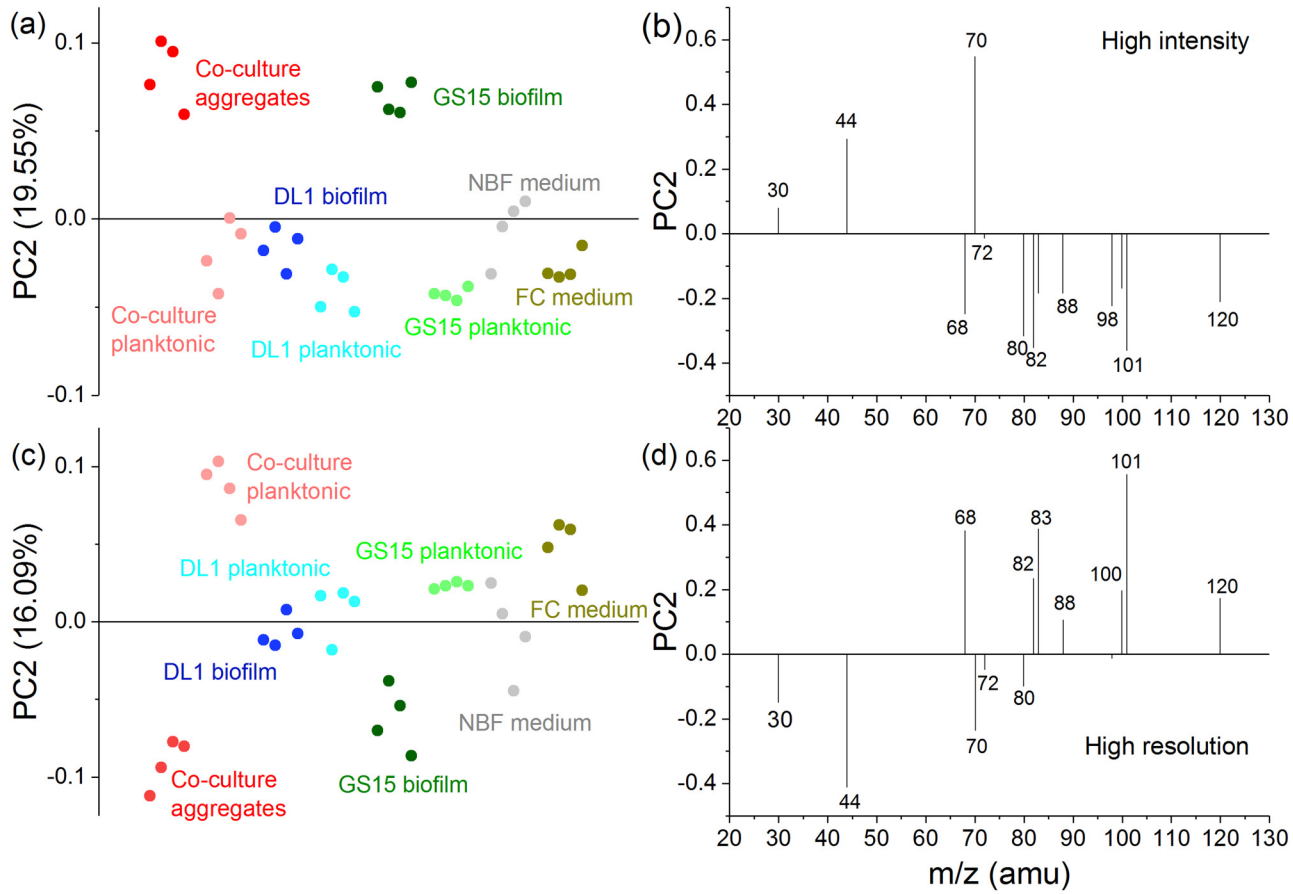


FIG. 5. Comparison of amino acid peak spectral PCA between high intensity and high resolution data in the positive mode. (a) PC2 scores plot of high intensity data; (b) PC2 loadings plot of high intensity data; (c) PC2 scores plot of high resolution data; and (d) PC2 loadings plot of high resolution data.

acid peaks are representative of proteins as a main EPS component due to ToF-SIMS ionization. High resolution data show better discrimination between coculture planktonic and coculture aggregates when only amino acid peaks are chosen.

D. Water clusters in relation to cocultured community and various controls

Water is an important and essential media for living activities of microbial communities and cells. Our previous report about *Shewanella oneidensis* MR-1 found, for the first time, a large amount of water clusters in living biofilms by using liquid SIMS.[21] In this work, we select water cluster peaks to compare spectral PCA analysis of high intensity data and high resolution data. Figure 6 depicts that high resolution data spectral PCA is more efficient to distinguish the aggregates and planktonic samples than high intensity data, especially for the coculture aggregates and coculture planktonic cultures, GS15 biofilm and GS15 planktonic cultures in PC2 in the positive mode. The FC medium shows a higher score in the PCA scores plots [Figs. 6(a) and 6(c)], suggesting a strong influence from the FC medium. For the high resolution data, PC2 spectral PCA analysis shows that water cluster peaks, such as m/z$^+$ 19 (H_2O) H^+, 145 $(H_2O)_8$ H^+, 199 $(H_2O)_{11}$ H^+, and 217 $(H_2O)_{12}$ H^+, dominate the FC medium, while other water cluster peaks, such as m/z$^+$ 37 $(H_2O)_2$ H^+, 91 $(H_2O)_5$ H^+, and 181 $(H_2O)_{10}$ H^+, are more representative of the NBF medium. Additionally, more water clusters are significant in the coculture aggregates samples, implying that the formation of the biofilm might change the distribution and relative abundance of water clusters. This is reasonable because biochemical reactions often occur in water; therefore, the water microenvironment is changed to better adapt to the living activities and the biofilm formation of the syntrophic community.[41,42]

Water cluster peak spectral PCA results of PC1 from high intensity and high resolution data are shown in Fig. S9 (Ref. 43). PC1 results do not discriminate coculture planktonic and coculture aggregates obviously, but similar loadings plots are obtained for both high intensity data and high resolution data. By comparing the spectral PCA of high intensity data and high resolution data using selected peaks from spectral overlay, solely amino acid peaks, and water cluster peaks, both high intensity data and high resolution data are useful for spectral analysis. High resolution data seem to perform a little better than

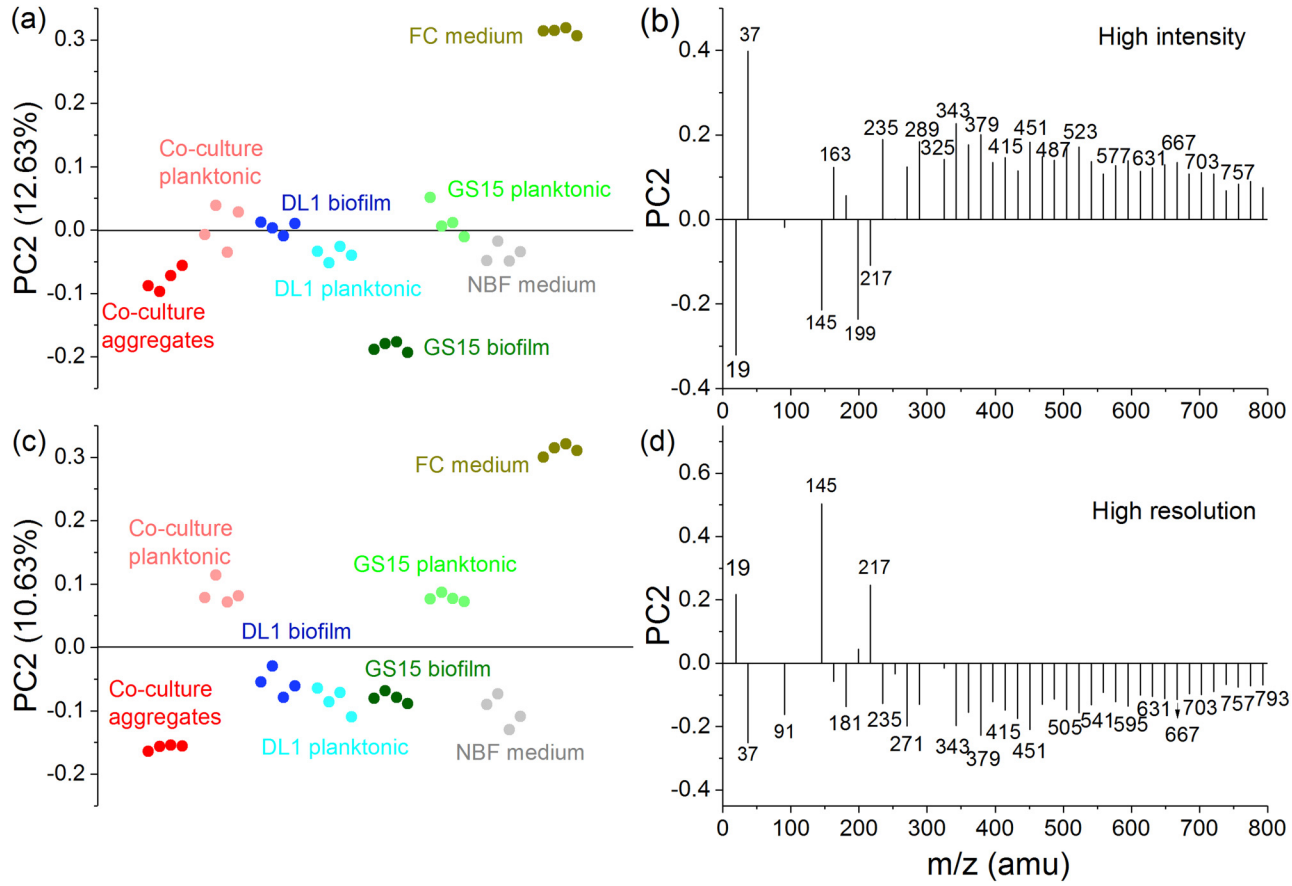


FIG. 6. Comparison of water cluster peak spectral PCA between high intensity and high resolution data in the positive mode. (a) PC2 scores plot of high intensity data; (b) PC2 loadings plot of high intensity data; (c) PC2 scores plot of high resolution data; and (d) PC2 loadings plot of high resolution data.

high intensity data for separating the coculture aggregates and coculture planktonic samples.

IV. SUMMARY AND CONCLUSIONS

In situ liquid SIMS has been used to reveal the surface chemical compositions of living biofilms. Particularly, syntrophic *Geobacter* species and single cultures are analyzed in this work. Our results show that peak selection is important for the multivariate spectral analysis and spectral overlay is proven as an effective method to reduce the matrix effect and reveal the main components in individual biofilms, cocultures, and planktonic cells. Amino acid and lipid peaks are the basis for biofilm formation and main components in extracellular secretions of *Geobacter*. Different amino acid and lipid peaks contribute to the discrimination of microbial samples in spectral PCA. In addition, peak selection provides an effective way to test and verify the hypothesis of microbial systems of interest. Amino acids as main components of EPS and water clusters as typical peaks in living biofilms are selected, respectively, for more spectral PCA. The results show that both high intensity and high resolution data are suitable for spectral analysis, but coculture biofilms and coplanktonic cultures seem to be a little better distinguished with high resolution data.

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